

START for ALL PARTICIPANTS

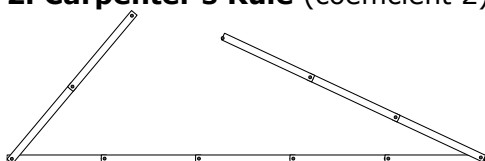
1. Matthew's Addition (coefficient 1)

$$\begin{array}{r} \spadesuit \quad \heartsuit \\ + \quad \spadesuit \quad \heartsuit \\ \hline = \quad \heartsuit \quad \diamondsuit \quad \clubsuit \end{array}$$

Matthew wrote an addition with digits that were all smaller than 6. He then stuck stickers over the digits: the same sticker on the same digit and different stickers on different digits.

What was the result of his addition?

2. Carpenter's Rule (coefficient 2)

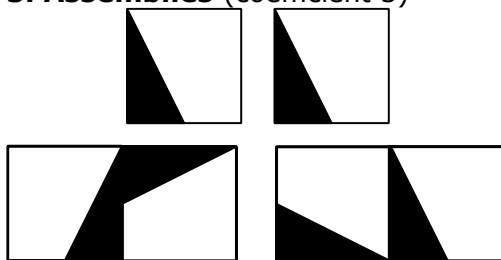


In his grandfather's workshop, Matthew found a carpenter's rule made up of 10 segments of 20 cm. He has fun making triangles, by folding it in two places.

How many different shapes of triangle can he get?

Note: two triangles are not different if we can put one on top of the other, possibly rotating or flipping one of the two. The folding in the drawing above would be an example of a flattened triangle: such triangles should not be counted.

3. Assemblies (coefficient 3)



example 1

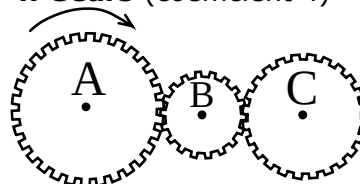
example 2

The two small squares at the top have a transparent part (in white) and a black part on the visible side and on the other side. We assemble these two squares to form a 2×1 rectangle as in the examples. You can turn these squares, and place them right side up or upside down. The black parts of the assembled rectangle must form one piece as in example 1; we will not allow the assemblies where the black parts touch at a single point as in example 2.

How many different rectangles can we get?

Note: if two rectangles are identical by turning one of them or placing one of them on the other side, they will only be counted for one figure.

4. Gears (coefficient 4)



Gear A has 30 teeth, gear B 18 teeth and gear C 25 teeth. Matilda spins gear A for exactly 10

complete rotations.

How many complete rotations will gear C make?

5. Matilda's Game (coefficient 5)

Matilda has in her hand a set of twenty-two cards numbered 0 to 21. She places a first card on the table, then she continues by placing cards one by one on the pile with the following rule: the numbers of two placed cards, one on top of the other must have at least one common digit. For example, if card #5 was on the table, she could only put card #15 on it.

When Matilda can no longer play, how many cards will there be in the pile, at most?

END for CE PARTICIPANTS

6. Complete the Multiplication (coefficient 6)

1 2 3 4 5 6 8

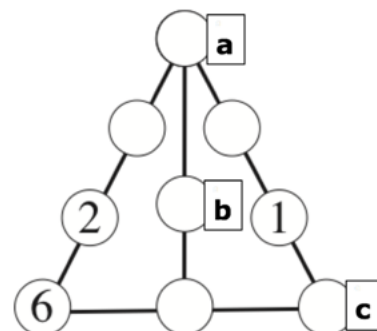
$$\begin{array}{r} \square \square \square \\ \times \quad 7 \\ \hline = \quad 9 \square \square \end{array}$$

Using five of the seven tokens shown at the top, complete the multiplication. **Which two tokens will not be used?**

Write the values of these tokens in order, from the smaller to the larger.

7. Triangle of the Year (coefficient 7)

Place the numbers 3, 4, 5, 7, 8 and 9 in the empty discs in the figure so that adding the numbers in any line always gives a result of 21. **Give the values in the discs a, b, c.**



8. Birthday (coefficient 8)

Max celebrated his birthday on November 20, 2020 which can be written as 20.11.20. He notices that this date is written with 3 different digits each written twice.

How many dates in 2020 were also written with 3 different digits each written twice, including 11.20.20?

Note: January 8, 2020 is written 08.01.20.

END for CM PARTICIPANTS

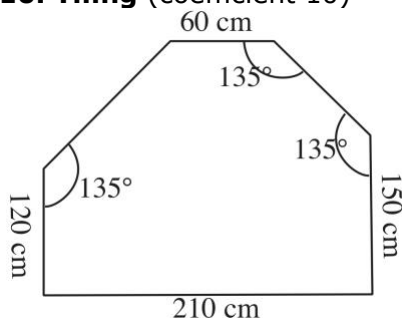
Problems 9 to 18: beware! For a problem to be completely solved, you must give both the number of solutions, AND give the solution if there is only one, or give any two correct solutions if there are more than one. For all problems that may have more than one solution, there is space for two answers on the answer sheet (but there may still be just one solution).

9. Black Squares (coefficient 9)

In a 15x15 square crossword puzzle, there are 32 black squares. In every column there are 1, 2 or 3 black squares: no column has 0 and four columns have only 1.

How many columns have 3 black squares?

10. Tiling (coefficient 10)



Basil wants to lay tiles in his entrance hall, sketched here. He will use square tiles of side of 30 cm. He can cut tiles and use the various pieces

to use as few tiles as possible.

How many tiles should he buy at least?

Note: the area of the joints is not counted.

11. The Zoo (coefficient 11)

Matilda and Matthew visit a zoo. In an enclosure, there are several Bactrian camels, which have two humps, several dromedaries, which have only one, and several donkeys, which have none. Matthew counted 60 hooves and Matilda counted 17 humps.

How many dromedaries does the enclosure have?

END for C1 PARTICIPANTS

12. A Remarkable Year (coefficient 12)

A given year has a year number written ABCD, the letters representing in order each digit in the year number. This year number is such that $ABCD + ABC + AB = 2021$

What is the year number?

13. The Jewel Box (coefficient 13)

For her birthday, Mathias gives his mother a jewel. The jewel is in a box in the shape of a rectangular parallelepiped, the dimensions of which are all whole numbers of cm. Two sides of the box are square and the other four are non-square rectangles. By calculating the volume of the box in cm^3 and its total area in cm^2 , Mathias notices that the two numbers obtained are equal.

What is the length in cm of the side of a square face of the box?

14. Augmentation Operation (coefficient 14)

$$12 \times 21 = 2021$$

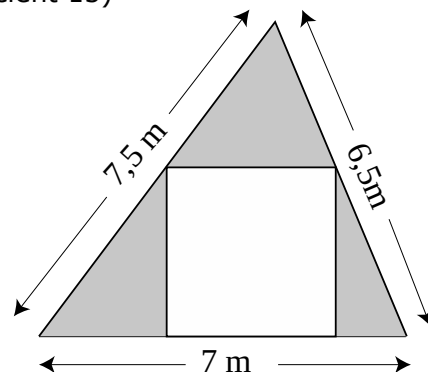
This multiplication is wrong!

What same number (positive or negative) must be added to each of the three numbers in the operation to make it correct?

END for C2 PARTICIPANTS

15. Attic (coefficient 15)

The drawing represents the cross-section of the roof of Matilda's house. Her parents arranged a bedroom under the roof so that the height of this room was equal to its



cross-sectional width, corresponding to the white square on the drawing.

What is this height or width?

Give the answer in metres, rounded to the nearest cm.

16. Signed Pyramid (coefficient 16)

-
+ -
- - +
+ - + +
- - + + +

This pyramid was built as follows:

- write five + or - signs on the lowest row
- above any two signs (in the middle), write the sign + or the sign - according to the "rule of signs" for the multiplication: + by + gives +, etc ...

If the bottom row contains 21 + or - signs, how many - signs can the entire pyramid contain, at most?

END for L1, GP PARTICIPANTS

17. So many triangles! (coefficient 17)

Matthew has embarked on a pharaonic project: to build all the (unflattened) triangles whose three angles measure whole numbers of degrees.

If he completes his project, how many dissimilar triangles (even if turned over) will he construct?

18. Duke Hunter's Wetland (coefficient 18)

Four cabins are on a perfectly circular road that surrounds a wetland.

The quadrilateral they form delimits Duke Hunter's shooting reserve.

The diagonals of the quadrilateral are perpendicular.

The total of the lengths of one pair of opposite sides of the quadrilateral is equal to 360 decametres (dam), while the total of the lengths of the other pair of opposite sides is equal to 450 dam.

The area of the shooting reserve is as large as possible.

What is the length of the longest side of the quadrilateral formed by the four cabins?

Give this length rounded to the nearest dam.

END for L2, HC PARTICIPANTS